

Building a graph

$$f(x) = 3^x$$

What points result for $x = -2, -1, 0, 1,$ and 2 ?

Growth graph

$$f(x) = 3^x$$

What are the domain, range, intercept, and horizontal asymptote?

Decay graph

$$g(x) = \left(\frac{1}{3}\right)^x$$

How does this graph differ from the graph of 3 to the power x?

Never zero

$$f(x) = 5^x$$

Can this graph intersect the x-axis?

End behavior

$$g(x) = \left(\frac{1}{4}\right)^x$$

What happens to $g(x)$ as x increases without bound?

Reflecting the input

$$g(x) = 2^{-x}$$

How does this graph relate to the graph of 2 to the power x?

Reflecting the output

$$g(x) = -2^x$$

What are the range and direction of change?

Vertical stretch

$$g(x) = 4 \cdot 2^x$$

How does every point on the graph of 2 to the power x change?

Horizontal shift

$$g(x) = 2^x - 3$$

Where does the point (0, 1) on the parent graph move?

Vertical shift

$$g(x) = 2^x + 3$$

What are the new range and horizontal asymptote?

Combining transformations

$$g(x) = 3 \cdot 2^{x+1} - 4$$

What transformations produce this graph from the parent graph?

Range after reflection and shift

$$g(x) = -3^x + 5$$

What is the range?

A transformed decay graph

$$g(x) = 2\left(\frac{1}{2}\right)^x - 3$$

What are the y-intercept and horizontal asymptote?

An x-intercept

$$g(x) = 2^{x-1} - 4$$

Where does this graph cross the x-axis?

Horizontal compression

$$g(x) = 2^{3x}$$

How does the point (1, 2) on the parent graph move?

Moving toward an asymptote

$$g(x) = 5 - 2^{-x}$$

What value does $g(x)$ approach as x increases without bound?

Reading graph information

The horizontal asymptote is $y = 3$, and the graph passes through $(0, 7)$.

$$g(x) = a \cdot 2^x + k$$

What values of a and k fit these graph features?

Sketch from a formula

$$g(x) = -2^{x-2} + 1$$

What sketch shows the intercepts, asymptote, and direction of change?

Function domain versus model domain

A is an account balance in dollars; t is years after opening the account.

$$A(t) = 300e^{0.08t}$$

How does the real-world domain differ from the formula domain?

Preparing for an inverse

$$f(x) = 2^x$$

If the graph is reflected across $y = x$, where do $(0, 1)$ and $(3, 8)$ move?